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Digital Product Passport - Application Programming Interfaces (APIs) for the product passport lifecycle management and searchability

Passeport numérique des produits - Interfaces de programmation d'applications (API) pour la gestion du cycle de vie et la recherchabilité du passeport d'un produit

Digitaler Produktpass - Programmierschnittstellen (APIs) für das Lebenszyklusmanagement und die Durchsuchbarkeit vom Produktpass

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European foreword

This document (EN 18222:2026) has been prepared by Technical Committee CEN/CLC/JTC 24 “Digital Product Passport – Framework and System”, the secretariat of which is held by DIN.

This European Standard shall be given the status of a national standard, either by publication of an identical text or by endorsement, at the latest by November 2026, and conflicting national standards shall be withdrawn at the latest by November 2026.

Attention is drawn to the possibility that some of the elements of this document may be the subject of patent rights. CEN shall not be held responsible for identifying any or all such patent rights.This document has been prepared under a standardization request addressed to CEN by the European Commission. The Standing Committee of the EFTA States subsequently approves these requests for its Member States.

For the relationship with EU Legislation, see informative Annex ZA, which is an integral part of this document.

Any feedback and questions on this document should be directed to the users’ national standards body. A complete listing of these bodies can be found on the CEN website.

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Introduction

0.1 Background

A digital product passport (DPP) is a key enabling mechanism to make product information traceable and accessible across value chains - supporting economic operators, manufacturers, distributors, repairers, recyclers and consumers to make informed decisions and to support a circular economy. The implementation of digital product passports will be carried out progressively. Sector-specific initiatives will determine the precise DPP content and requirements for individual product groups.

To support the implementation of DPPs, 8 standards have been developed so far:

— EN 18219:2026 – Digital product passport – Unique identifiers

— EN 18220:2026 – Digital product passport – Data carriers

— EN 18216:2026 – Digital product passport – Data exchange protocols

— EN 18222: 2026 – Digital Product Passport – Application Programming Interfaces (APIs) for the product passport lifecycle management and searchability (this document)

— EN 18223:2026 – Digital Product Passport – System interoperability

— EN 18221:2026 – Digital product passport – data storage, archiving, and data persistence

— EN 18239:—[1] – Digital Product Passport – access rights management, information system security, and business confidentiality

— EN 18246:—[2] – Digital Product Passport – Data authentication, reliability and integrity

0.2 Overview

The Ecodesign for Sustainable Products Regulation (ESPR) is an initiative by the European Commission aimed at promoting sustainable products by setting comprehensive requirements for product design, production, and lifecycle management. Central to this initiative is the digital product passport (DPP), which tracks and provides essential information about a product's sustainability attributes. Beside of that also other data can be stored in a DPP depending on future delegated acts and the needs of economic operators.

This document introduces the specifications for the API of the DPP, as mandated by the ESPR. The API is designed to enhance the searchability of DPPs and to support interactions throughout the lifecycle of a product's DPP. Furthermore, it also provides an API to register a DPP at the DPP registry.

The API methods are presented on a technology-neutral level, detailing the expected inputs and outputs for each method. A detailed technological implementation using a REST-HTTP API is described in the Clause 8, providing guidelines for developers to implement the functionality effectively within their specific environments.

The specification document of EN 18223 describes the logical content of the payload that is used by the API. EN 18216 defines the basic principles of the exchange protocol and the allowed serialization formats of the payload data. Security requirements need to be followed based on the specification document of EN 18246[2].

[1] Under preparation. Stage at the time of publication: prEN 18239:2025.
[2] Under preparation. Stage at the time of publication: prEN 18246:2025.

1 Scope

This document aims to standardize the specifications for the API of the digital product passport (DPP) as mandated by the ESPR of the European Commission. The purpose of this API is to facilitate the searchability of DPPs, as well as to provide the necessary means for interactions throughout the lifecycle of a product's DPP.

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

EN 18221, *Digital product passport - data storage, archiving, and data persistence*

3 Terms and definitions

For the purposes of this document, the following terms and definitions apply.

ISO and IEC maintain terminology databases for use in standardization at the following addresses:

— ISO Online browsing platform: available at http://www.iso.org/obp

— IEC Electropedia: available at http://www.electropedia.org/

3.1
application programming interface
API
set of *methods* (3.3) provided by an application component for use by other application components

3.1.1
representational state transfer application programming interface
REST API
service that allows for interaction with resources via a stateless, client-server architecture, typically using standard HTTP calls like GET, POST, PUT, and DELETE to perform *methods* (3.3) by an application component on these resources

3.2
digital product passport
DPP
digital record of product characteristics throughout its life cycle

Note 1 to entry: Example characteristics include environmental sustainability, environmental impact, and recyclability

3.3
method
particular way used to perform a specific action on a resource

EXAMPLE read a DPP

3.4
parameter
data provided by a client, that is needed to perform a *method* (3.3)

EXAMPLE DPP ID for read DPP

3.5
result
return value after the execution of a *method* (3.3)

EXAMPLE the DPP

3.6
unique registration identifier
identifier returned after uploading the unique identifiers and any other data required to the DPP registry

EXAMPLE Unique identifiers can be unique product identifier, unique facility identifier, unique operator identifier of the economic operator etc.

4 Specification of the Life Cycle API (Main Methods)

4.1 General

In this clause only the API methods as can be made available by the custodian of a DPP (either economic operator or service provider) are contained. These methods are specified in an abstract manner, the precise implementation is provided in Clause 8, which also contains information how the input and output parameters are provided.

The following methods shall be made available by the creator of the digital product passport or their main digital product passport service provider as well as by back-up digital product passport service provider:

— ReadDPPById

— ReadDPPByProductId

— ReadDPPIdsByProductIds

The following methods should be made available by the creator of the digital product passport or their main digital product passport service provider as well as by back-up digital product passport service provider:

— ReadDataElement

The following methods should be made available by the main digital product passport service provider as well as by back-up digital product passport service provider:

— CreateDPP

— DeleteDPPById

— ReadDPPVersionByIdAndDate

The following methods shall be made available by the creator of the digital product passport or their main digital product passport service provider if authorized third parties have access rights to update parts of the DPP:

— UpdateDPPById

The following methods should be made available by the creator of the digital product passport or their main digital product passport service provider if authorized third parties have access rights to update parts of the DPP:

— UpdateDataElement

In principle, the DPP API will offer these set of methods, that is requested with some parameters, performing a behaviour (e.g. executing an action), and responding with a result. These expectations are formally described for each method in the following subclauses.

Following methods support querying that may be used to create search functionality

— ReadDPPByProductId

— ReadDPPIdsByProductIds

— ReadDPPVersionByProductIdAndDate

Content negotiation according to EN 18216 [1] should be used if multiple response types are supported.

4.2 Method ReadDPPById

This method returns the DPP through a known DPP identifier. See Table 1.

Access rules define the data elements of DPP to be returned in the response. For details see EN 18246[2] [2].

Table 1 — ReadDPPById

Method Name	ReadDPPById			
Explanation	Returns the DPP with the specified DPP ID			
Name	**Description**	**Mandatory**	**Type**	**Cardinality**
Input Parameter				
digitalProductPassportId	The DPP's unique ID	yes	Identifier	1
Output Parameter				
statusCode	Response status code	yes	StatusCode	1
payload	Requested DPP	yes	DigitalProductPassport	0..1

4.3 Method ReadDPPByProductId

This method returns the current active DPP (latest version) through the unique product identifier as defined in EN 18219 [3]. See Table 2.

Access rules define the data elements of DPP to be returned in the response. For details see EN 18246[2] [2].

Table 2 — ReadDPPByProductId

Method Name	ReadDPPByProductId			
Explanation	Returns the current active DPP with the specified Product ID			
Name	**Description**	**Mandatory**	**Type**	**Cardinality**
Input Parameter				
productId	Product ID Unique product identifier as defined in EN 18219 [3]	yes	Identifier	1
Output Parameter				
statusCode	Response status code	yes	StatusCode	1
payload	Requested DPP	yes	DigitalProductPassport	0..1

4.4 Method ReadDPPVersionByIdAndDate

This method returns a DPP based on its digital product passport identifier and given date. The method is optional for the economic operator. See Table 3.

Access rules define the data elements of DPP to be returned in the response. For details see EN 18246[2] [2].

Table 3 — ReadDPPVersionByIdAndDate

Method Name	ReadDPPVersionByIdAndDate			
Explanation	Returns a DPP version based on its digitalProductPassportId and given date			
Name	**Description**	**Mandatory**	**Type**	**Cardinality**
Input Parameter				
digitalProductPassportId	DPP ID	yes	Identifier	1
date	Date value for which the DPP is requested	yes	Timestamp (UTC-based)	1
Output Parameter				
statusCode	Response status code	yes	StatusCode	1
payload	Requested DPP	yes	DigitalProductPassport	0..1

Dateformat see ISO 8601-1 [4] and EN 18223 [5].

4.5 Method ReadDPPIdsByProductIds

This method returns a list of DPP identifiers that matches a set of product identifiers.

The input parameters “limit” and “cursor” can be set by the client to control pagination. The value of the cursor shall not be empty. See Table 4.

Table 4 — ReadDPPIdsByProductIds

Method Name	ReadDPPIdsByProductIds			
Explanation	Returns a list of DPP identifiers based on discovery information			
Name	**Description**	**Mandatory**	**Type**	**Cardinality**
Input Parameter				
productId	Product IDs Unique product identifiers as defined in EN 18219 [3]	yes	Identifier[]	1..*
Output Parameter				
statusCode	Response status code	yes	StatusCode	1
payload	DPP Identifiers of all DPPs matching the discovery information	yes	Identifier[]	0..*

Identifier[] is a placeholder for an array.

4.6 Method CreateDPP

Method to create a new DPP. Access rules control whether the creation is allowed or not. For details see EN 18246[2] [2]. It returns the ID of the DPP. See Table 5.

Table 5 — CreateDPP

Method Name	CreateDPP			
Explanation	Creates DPP			
Name	**Description**	**Mandatory**	**Type**	**Cardinality**
Input Parameter				
dpp	DPP	yes	DigitalProductPassport	1
Output Parameter				
statusCode	Response status code	yes	StatusCode	1
digitalProductPassportId	DPP ID	yes	Identifier	0..1

4.7 Method UpdateDPPById

Updates the content of the DPP. The submitted DPP only contains data that need to be updated or are extended with new data. It can contain a single data element or can be a set of different element information that needs to be updated. See Table 6.

Access rules control whether the patch is allowed or not. For details see EN 18246[2] [2].

If the update of some parts fails the complete update process will fail and there should be no changes adopted in the DPP.

All changes to the digital product passport shall be archived in accordance with EN 18221.

Table 6 — UpdateDPPById

Method Name	UpdateDPPById			
Explanation	Partial update of a DPP with a specified DPP ID			
Name	**Description**	**Mandatory**	**Type**	**Cardinality**
Input Parameter				
digitalProductPassportId	DPP identifier	yes	Identifier	1
partialDPP	DigitalProductPassport (only contains parts that needs to be updated)	yes	DigitalProductPassport	1
Output Parameter				
statusCode	Response status code (should reflect the information when the update fails)	yes	StatusCode	1
payload	Updated DPP	yes	DigitalProductPassport	0..1

4.8 Method DeleteDPPById

Method that removes a DPP of a specific DPP ID. This gets relevant for the end of life of a DPP. Access rules control whether deletion of DPPs is allowed or not. For details see EN 18246[2] [2]. See Table 7.

Table 7 — DeleteDPPById

Method Name	DeleteDPPById			
Explanation	Deletes DPP with the specified DPP ID			
Name	**Description**	**Mandatory**	**Type**	**Cardinality**
Input Parameter				
digitalProductPassportId	DPP identifier	yes	Identifier	1
Output Parameter				
statusCode	Response status code	yes	StatusCode	1

5 Specification of the DPP Registry API for the Registration Process

5.1 General

This clause covers access to external methods of the DPP registry.

The DPP registry shall provide the following method:

- RegisterProductDPP

5.2 Method RegisterProductDPP

This method enables the registration of a new DPP at the DPP registry. This method is served by the DPP registry server. It returns a unique registry identifier at the requesting client. See Table 8.

Table 8 — RegisterProductDPP

Method Name	RegisterProductDPP			
Explanation	Register a new DPP at the registry			
Name	**Description**	**Mandatory**	**Type**	**Cardinality**
Input Parameter				
dppRegistryEntry	Contains information relevant for registration	yes	DppRegistryEntry	1
Output Parameter				
statusCode	Response status code	yes	StatusCode	1
registrationId	unique registration identifier	yes	String	0..1

6 Specification of the Fine Granular API Methods of the Life Cycle API

6.1 General

The operations in this clause may be made available by custodians of DPPs or archiving services.

6.2 Method ReadDataElement

This method allows to return a specific data element by its unique identifier path of a specific DPP.

See Table 9.

Table 9 — ReadDataElement

Method Name	ReadDataElement			
Explanation	Returns a specific data element by providing the absolute path of the data element			
Name	**Description**	**Mandatory**	**Type**	**Cardinality**
Input Parameter				
digitalProductPassportId	DPP identifier	yes	Identifier	1
elementIdPath	Path to the specific data element	yes	String	1
Output Parameter				
statusCode	Response status code	yes	StatusCode	1
payload	Requested data element	yes	DataElement	1

6.3 Method UpdateDataElement

This method updates a specific data element of a DPP. The method also allows to amend or remove data element information. See Table 10.

Table 10 — UpdateDataElement

Method Name	UpdateDataElement			
Explanation	Update of a data element within a specific DPP.			
Name	**Description**	**Mandatory**	**Type**	**Cardinality**
Input Parameter				
digitalProductPassportId	DPP identifier of the DPP where the specific data element needs to be updated	yes	Identifier	1
elementIdPath	Path to the specific data element	yes	String	1
payload	Content of data that needs to be updated	yes	DataElement	1
Output Parameter				
statusCode	Response status code	yes	StatusCode	1
payload	Changed data element	yes	DataElement	1

7 Parameter types

7.1 Data types

Table 11 below provides information on the types used for life cycle, registration, and fine granular API methods.

Table 11 — Data types

Type	Explanation
Identifier	Identifier as **String** as defined in EN 18223
StatusCode	StatusCode based on Table 15 in Clause 7
DppRegistryEntry	Digital product passport header as **Object** including key-value pairs: — uniqueProductIdentifier as **String** — digitalProductPassportId as **String** — uniqueEconomicOperatorIdentifier as **String** — uniqueEconomicOperatorIdentifier of the Back-up operator as **String** — dppApiEnd point as **Uri**
DigitalProductPassport	Digital product passport as **Object** based on EN 18223 as well as any DataElement subsets of the product DigitalProductPassport to manage updates
Timestamp	dateTime format based on ISO 8601-1 according to EN 18223
DataElement	dataElement defined in EN 18223

7.2 Status code, error handling and result messages

In case of a failed method execution, a result object should be returned containing more information about the reasons why the method failed to execute.

In case of a failed request, response may only include error object.

See Table 12 to Table 15.

Table 12 — General result object

Class Name	*Result*		
Explanation	The result object		
Inherits from	—		
semanticId	https://jtc24/dpp/API/DataTypes/Result		
Attribute	**Explanation**	**Type**	**Cardinality**
message	Additional message containing information for the requester	Message	0..*

Table 13 defines the structure of the message in the general result object.

Table 13 — Message structure

Class Name	*Message*		
Explanation	A message containing more information for the requester about a certain happening in the backend		
Inherits from	—		
Attribute	**Explanation**	**Type**	**Cardinality**
messageType	The message type	MessageTypeEnum	1
text	The message text as defined in Table 15	String	1
code	Technology-dependent status or error code as defined in Table 15	String	0..1
correlationId	Identifier to relate several result messages throughout several systems	String	0..1
timestamp	Timestamp of the message	DateTime	0..1

Table 14 defines the value range for the message type in a message.

Table 14 — MessageTypeEnum

Enumeration	*MessageTypeEnum*
Explanation	The message type
Literal	**Explanation**
Info	Used to inform the user about a certain fact
Warning	Used for warnings; warnings can lead to errors in the subsequent execution
Error	Used for handling errors
Exception	Used in case of an internal and/or unhandled exception

Table 15 defines the allowed values for the status code in a message.

Table 15 — Status code of response messages

Generic Status Code	Meaning	Has Result Object	HTTP Status Code
Success	Success	No	200
SuccessCreated	Successful creation of a new resource	No	201
SuccessAccepted	The reception of the request was successful	No	202
SuccessNoContent	Success with explicitly no content in the payload	No	204
ClientErrorBadRequest	Bad or malformed request	Yes	400
ClientNotAuthorized	Wrong or missing authorization credentials	Yes	401
ClientForbidden	Authorization has been refused	Yes	403
ClientMethodNotAllowed	Method request is not allowed	Yes	405
ClientErrorResourceNotFound	Resource not found	Yes	404
ClientResourceConflict	Conflict-creating resource (resource already exists)	Yes	409
ServerInternalError	Unexpected error	Yes	500
ServerNotImplemented	The server does not support the functionality to fulfil the request	No	501
ServerErrorBadGateway	Bad gateway	Yes	502

8 Mappings

8.1 General

This clause specifies the mapping of the logical model as defined in Clause 4, Clause 5 and Clause 6 to HTTPS/REST.

Table 16, Table 17 and Table 18 show the corresponding logical method names corresponding to Clause 4, Clause 5 and Clause 6, the associated HTTP methods, the REST-Resource Path, if the input parameter are provided in-path of the resource path or as payload body, as well the expected result.

For update operations [5] RFC 7396 can be considered for implementation.

The Version-Prefix (“v1/”) in REST-Path shall be replaced for further versions accordingly.

The value of elementIdPath shall follow [6] RFC 9535 JSONPath.

All in-path parameters like dppId should be escaped (percent encoding) within Urls.

An optional “representation” query flag for the GET, POST, and PATCH methods is used that can take two values: “compressed” (default) and “full“.

The “compressed“ value indicates that the client is requesting the payload containing the compressed DPP-based representation (see 5.2 in EN 18223).

The “full“ value will indicate that the client is requesting the payload containing the full DPP-based representation (see Annex A and Annex B in EN 18223).

If the “representation“ query is not used, the “compressed“ representation is assumed in the payload.

8.2 HTTPS/REST for Life Cycle API

Table 16 provides the core HTTPS methods of the DPP life cycle API.

Table 16 — Life Cycle API

Method Name	**HTTP Method**	**REST-Path**	**In-path /Query Parameter**	**Request Body**	**Result** (except status code and pagination)
ReadDPPById	GET	v1/dpps/{dppId}	DPP ID	–	DPP
ReadDPPByProductId	GET	v1/dppsByProductId/{productId}	Product ID	–	DPP (latest DPP version)
ReadDPPVersionByIdAndDate	GET	v1/dppsByIdAndDate/{dppId}?date = {timestamp}	DigitalProductPassportId (as in-path) and Timestamp (as query)	–	DPP (based on the given date)
ReadDPPIdsByProductIds	POST	v1/dppsByProductIds	–	Set of Product IDs	Set of DPP IDs
CreateDPP	POST	v1/dpps	–	DPP	DPP ID
UpdateDPPById	PATCH	v1/dpps/{dppId}	DPP ID	PartialDPP	DPP (updated)
DeleteDPPById	DELETE	v1/dpps/{dppId}	DPP ID	–	–

8.3 HTTPS/REST for Register API for Register

Table 17 provides the HTTPS method for the Register API that needs to be called to register a DPP.

Table 17 — Register API for Register

Method Name	**HTTP Method**	**REST-Path**	**In-path Parameter**	**Request Body**	**Result** (except status code and pagination)
RegisterProductDPP	POST	v1/registerDPP	-	dppRegistryEntry object	Registration ID

8.4 HTTPS/REST for Fine Granular Life Cycle API

Table 18 — Fine Granular Life Cycle API

Method Name	**HTTP Method**	**REST-Path**	**In-path Parameter**	**Request Body**	**Result** (except status code and pagination)
ReadDataElement	GET	v1/dpps/{dppId}/elements/{elementIdPath}	DPP ID (dppId) and unique path to the element		DataElement
UpdateDataElement	PATCH	v1/dpps/{dppId}/elements/{elementIdPath}	DPP ID (dppId) and unique path to the element (elementPath)	Data element that needs to be updated	DataElement (updated)

Annex ZA
(informative)

Relationship between this European Standard and the essential requirements of (EU) Regulation 2024/1781 aimed to be covered

This European Standard has been prepared under a Commission's standardization request M/604 Amd 1 to provide one voluntary means of conforming to essential requirements of (EU) Regulation 2024/1781 of 13 June 2024 of the European Parliament and of the Council of the European Union establishing a framework for the setting of ecodesign requirements for sustainable products, amending Directive (EU) 2020/1828 and Regulation (EU) 2023/1542 and repealing Directive 2009/125/EC.

Once this standard is cited in the Official Journal of the European Union under that Regulation, compliance with the normative clauses of this standard given in Table ZA.1 confers, within the limits of the scope of this standard, a presumption of conformity with the corresponding essential requirements of that Regulation and associated EFTA regulations.

Table ZA.1 — Correspondence between this European Standard and Articles of (EU) Regulation 2024/1781

Essential requirements of (EU) Regulation 2024/1781	Clause(s)/subclause(s) of this EN	Remarks/Notes
Article 10.1.d	4.5	
Article 10.1.f	4.5	
Article 10.3.a	4.5	
Article 10.3.b	4.5	
Article 11.a	4, 4.5	The presumption of conformity only applies to technical interoperability of end-to-end communication and data transfer
Article 11.b	4.5	

WARNING 1 — Presumption of conformity stays valid only as long as a reference to this European Standard is maintained in the list published in the Official Journal of the European Union. Users of this standard should consult frequently the latest list published in the Official Journal of the European Union.

WARNING 2 — Other Union legislation may be applicable to the falling within the scope of this standard.